

RESEARCH

Open Access



An enhanced deep learning framework for DeepFake detection using EfficientNet-B3 comparative evaluation of deep and machine learning techniques

B. G. Deepa^{1*}, C. K. Lokesh², D. Umamaheswari¹, B. Ayshwarya³, P. V. Yethish² and K. P. Suhaas⁴

*Correspondence:

B. G. Deepa

deepabg03@gmail.com

¹Department of Computer Science,
Christ University, Bangalore, India

²School of Computer Science
and Applications, REVA University,
Bangalore, India

³Department of Computer Science,
Kristu Jayanti Deemed to Be
University, Bangalore, India

⁴Department of Information
Science and Engineering, The
National Institute of Engineering,
Mysuru, India

Abstract

Over the last few years, spread of DeepFake content has become major issue on digital platforms, calling for strong detection systems. This research investigates and compares the performance of machine learning (ML) techniques and the deep learning (DL) techniques for DeepFake image categorization. The ML technologies like SVM, Random Forest, and ensemble models were utilized based on hand-designed features like GLCM and HOG, whereas DL techniques utilized strong architectures like EfficientNet-B3 and InceptionV3. The research also investigates effect of datasets of different size on model's performance in order to analyze scalability and generalization better. Experimental findings exhibit the higher performance of DL models in accuracy, with EfficientNet-B3 obtaining the highest value of 97.95%, while ML models provided computational efficiency and interpretability. This work adds a comparative view on the practical merits and demerits of both paradigms on the problem of DeepFake detection. In addition to benchmark datasets, the model was further evaluated on AI-generated images from ChatGPT and Gemini, successfully detecting them as fake, demonstrating strong generalization.

Keywords DeepFake detection, EfficientNet-B3, Convolutional neural networks (CNN), Machine learning models, Feature extraction, Transfer learning, Image forensics, Ensemble classifiers, Comparative performance evaluation, Fake Image Identification, Deep learning framework, Machine learning, Neural Network Architecture Optimization, Digital data authentication, Binary classification

1 Introduction

Advancements in generative Modeling Techniques, especially via Generative Adversarial Networks (GANs), has led to very realistic synthetic media, also referred to as DeepFakes. Although such technologies have facilitated enormous innovations in entertainment, visual effects, and data augmentation, they also present important risks to digital trust, privacy, and information integrity. DeepFakes have been used more and more for



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

nefarious reasons, such as political manipulation, celebrity impersonation, and misinformation campaigns, rendering their detection a very important area of research.

DeepFake detection can be considered as a binary division task, with the objective of distinguishing between authentic and synthetically altered visual data. To address this challenge, Machine Learning (ML) and Deep Learning (DL) techniques have been widely explored. Traditional Machine Learning methods typically rely on hand-engineered features such as the Gray Level Co-occurrence Matrix (GLCM) and Histogram of Oriented Gradients (HOG), followed by the classifiers such as support vector machines (SVMs) and the Random Forests. While these techniques are computationally efficient and offer interpretability, their performance is heavily dependent on the quality and relevance of the extracted features.

Conversely, DL models particularly Convolutional Neural Networks (CNNs) have demonstrated the ability to discover intricate patterns and hierarchies clearly from image data. Cutting-edge architectures like EfficientNet-B3, InceptionV3, fine-tuned using transfer learning,

exhibit outstanding performance in image classification tasks, including detecting DeepFakes. While DL models offer significant benefits, they generally require bigger datasets and greater computing capabilities to operate optimally.

This research conducts a comparative study of different ML and DL models deployed on the DeepFake image classification task. Of particular interest is assessing how the models scale across different dataset sizes, from 1000 to 5000 images per class, in terms of their robustness and scalability. The “DeepFake and Real Images” Kaggle dataset served as the baseline dataset for all experiments.

By careful experimentation and analysis, this research seeks to illuminate the accuracy, computational complexity, and sensitivity to data quantity trade-offs among conventional ML and advanced DL techniques. This finding provides valuable insights into model selection methodologies for actual DeepFake detection use cases.

While most studies evaluate solely on standard benchmark datasets, we further assessed our framework on images generated by state-of-the-art AI tools such as ChatGPT and Gemini, demonstrating its ability to generalize to emerging generative content.

2 Significance and contribution

Deepfake’s emergence has introduced a new digital era of deception that threatens the veracity of visual content in the social media, journalism, and security arenas. With more sophisticated and ubiquitous manipulated media, the need for powerful detection mechanisms becomes increasingly critical. This work fills this requirement by conducting a comprehensive comparison between the Machine Learning (ML) and Deep Learning (DL) approaches for the task of DeepFake image classification. The contribution of this work is twofold: comparing classical ML methods with current advanced DL models and examining how the performance changes with varying dataset sizes.

One of the main contributions of this research is the extensive benchmarking of several ML and DL models in a unified experimental framework. Machine Learning models, such as the Support Vector Machine, Random Forest, and ensemble techniques are assessed based on several hand-engineered features, including the Gray Level Co-occurrence Matrix (GLCM), Histogram of oriented Gradients (HOG), the color-edge descriptors. These models are also contrasted against DL architectures including

EfficientNet-B3 and InceptionV3 that apply transfer learning and fine-tuning to pick out intricate features from image data. This contrast provides a direct focus on trade-offs between interpretability, accuracy, and computational efficiency.

Another interesting feature of this work is its investigation of dataset size sensitivity. Models were trained and tested on different volumes of data—between 1000 and 5000 images per class—on the publicly available "DeepFake and Real Images" Kaggle dataset. This is an interesting insight into how model performance varies with the amount of data available, which informs practitioners in the selection of models depending on the size and quality of their datasets.

Moreover, the work prioritizes reproducibility and applicability. By using a publicly available dataset and common architectures, this work makes it straightforward for others to replicate and extend the experimental setup. The results also show that, although DL models are substantially better than ML models in pure accuracy (EfficientNet-B3 having a maximum of 98.08% accuracy), ML methods remain competitive when there is a restriction on computational resources or interpretability is desired.

In essence, this work is both theoretically and practically impactful by contrasting different learning paradigms, providing insights into dataset utilization, and pushing the existing state of DeepFake detection in a scalable, reproducible, and interpretable manner.

3 Review of literature

Khan et al. [1] proposed an ensemble-based framework utilizing VGG16, InceptionV3, and XceptionNet, achieving a detection accuracy of 96.5%. Their fusion strategy enhanced robustness by combining diverse CNN architectures. Wodajo et al. [2] developed GenConViT, a hybrid of ConvNeXt and Swin Transformers, which reached 95.8% accuracy and an AUC of 99.3%, effectively capturing spatial and contextual inconsistencies.

Coccomini et al. [3] integrated EfficientNet with the Vision Transformers for the DeepFake video discovery, achieving an AUC value of 0.951 and the high F1 score of 88.0% without relying on ensembles, showcasing a minimal yet effective fusion strategy. Thing [4] conducted a comparative study between CNNs and Transformers and reported accuracy as high as 99.73%, suggesting model selection should depend on dataset characteristics.

Sun et al. [5] proposed a cross-domain feature fusion method using edge and semantic features, reaching 90.5% accuracy and showing improved generalization across datasets. Qiu et al. [6] introduced D2Fusion, a dual-domain fusion framework that combines spatial and frequency domain features using bi-directional attention mechanisms. This method effectively captures both the local and the global forgery artifacts, achieving superior performance across multiple DeepFake datasets.

In recent studies, researchers have investigated richer feature spaces. Qiu et al. [6] proposed D2Fusion, a dual-domain fusion framework that integrates spatial and frequency features with the help of bidirectional attention mechanisms. This method effectively captured both local and global artifacts and achieved high detection scores across multiple datasets.

Chen et al. [7] projected DefakeHop, a lightweight yet high-performance DeepFake indicator based on successive subspace learning. Utilizing the c/w Saab transform for

feature extraction, DefakeHop achieved an AUC of 100% on the UADFV dataset, demonstrating its efficacy with minimal computational resources.

Beyond approaches that rely on feature fusion and lightweight design, several studies have investigated other important aspects such as improved training schemes, attention-driven mechanisms, and cues specific to video data. Wodajo and Atnafu [8] explored convolutional vision transformers tailored for video-based forgery detection, whereas Zhao et al. [9] proposed a self-consistency learning strategy that enhances resistance to subtle manipulations by enforcing agreement across frame-level representations. Comprehensive surveys by Passos et al. [10] and Heidari et al. [14] summarized recent progress and pointed to promising directions like model efficiency and stronger data augmentation methods. In addition, Badale et al. [15] explored early neural-network-based approaches for deepfake detection, contributing foundational insights into how basic deep learning architectures can separate authentic from manipulated media. Tran et al. [11] designed a CNN framework with attention modules focused on regions most likely to contain tampering, while Rahman [12] demonstrated that CNN and RCNN models can successfully capture localized inconsistencies in forged media. Bondi et al. [13] further showed that thoughtful training procedures and augmentation techniques play a critical role in building more robust detection systems.

The surveyed literature illustrates the rapid evolution and diversity of DeepFake detection strategies. From CNN ensembles and hybrid CNN-transformer models to frequency-domain feature fusion scholars have adopted a expansive range of techniques to address the growing complexity of synthetic media. Ensemble-based models like those by Khan et al., and hybrid architectures such as GenConViT and EfficientNet-ViT, showcase strong accuracy through architectural fusion. Together, these studies emphasize that high-performing detection models often fuse multiple feature domains, incorporate attention mechanisms, or adopt transformer-based backbones to capture diverse cues in manipulated media. However, many of these approaches are computationally heavy or oriented toward video analysis. In contrast, the present work builds on these insights by leveraging a lightweight yet powerful architecture, EfficientNet-B3, to deliver high-accuracy detection in an image-based setting while maintaining scalability and computational efficiency.

4 Dataset and preprocessing

This study utilized the publicly available DeepFake and Real Images dataset from Kaggle repository [<https://www.kaggle.com/datasets/manjilkarki/deepfake-and-real-images>], which contains a collection of synthetic and authentic facial images as shown in Figs. 1 and 2. Additional AI-generated image samples used for validation are available from the corresponding author on reasonable request.

The dataset is organized into separate folders for training, validation, and testing, each comprising real and fake image subdirectories. For experimental consistency, a balanced subset of up to 5,000 real and 5,000 fake images was selected for model training and evaluation.

To enhance model generalization and mitigate overfitting, a comprehensive data preprocessing pipeline was applied. All images were resized to 224×224 pixels, followed by normalization using standard ImageNet mean and standard deviation values. Data augmentation techniques including random horizontal flipping, rotation (up to 30 degrees)

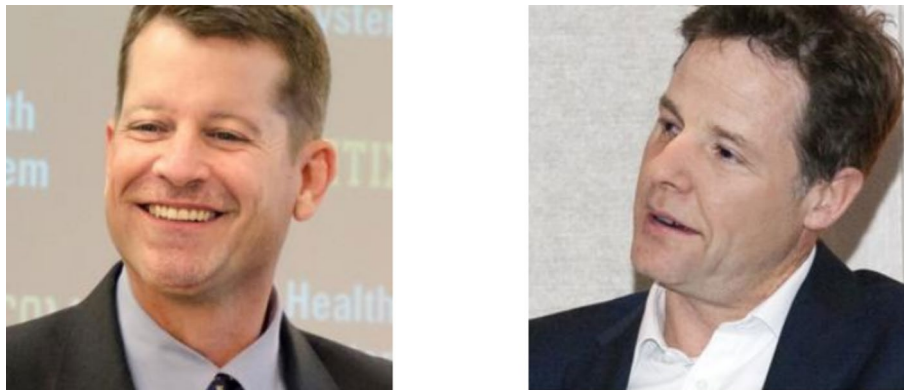


Fig. 1 Fabricated images



Fig. 2 Real images

were applied to the training images to increase variability and robustness. Testing and validation images were only resized and normalized to preserve evaluation consistency.

A custom PyTorch dataset class was implemented to load and label real and fake images dynamically. The full dataset was further split into 80% for guiding and 20% for internal examining using a randomized approach. Data was loaded in batches of 8 using PyTorch's DataLoader to optimize memory usage and computational efficiency during training.

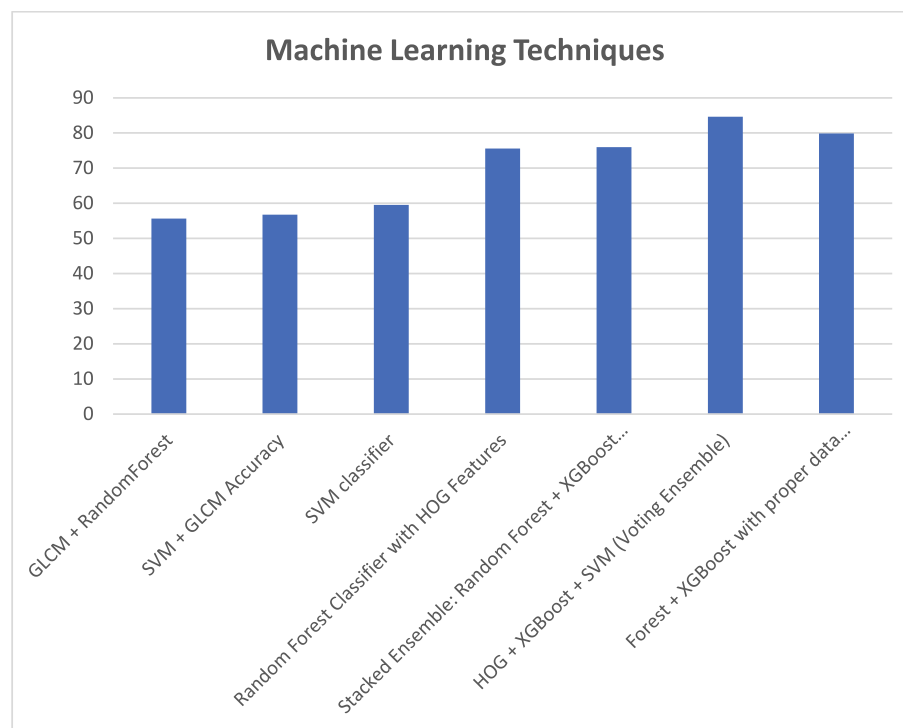
All deep learning experiments were conducted using the EfficientNet-B3 model, leveraging the timm library for model instantiation with pretrained weights. The model's final classification layer was replaced with a custom binary classification head using Sigmoid activation. Training was performed on GPU (if available), using the adam optimizer and the binary cross-entropy loss function. The framework underwent training for 10 epochs, and best-performing weights (based on training accuracy) were saved for final evaluation. Performance metrics for instance accuracy, confusion matrix, and classification report was employed to evaluate the model's performance.

5 Result analysis

This section presents the examination of numerous Machine Learning (ML) and the Deep Learning (DL) models for deepfake detection, based on their execution with respect to accuracy, computational efficiency, feature selection strategies.

Table 1 Machine learning techniques

Machine leaning techniques	
GLCM + RandomForest	55.63
SVM + GLCM Accuracy	56.8
SVM classifier	59.53
Random Forest Classifier with HOG Features	75.58
Stacked Ensemble: Random Forest + XGBoost + SVM (Meta: Logistic Regression)	76
HOG + XGBoost + SVM (Voting Ensemble)	84.62
Forest + XGBoost with proper data preprocessing, HOG feature extraction, PCA, and hyperparameter tuning	79.87

**Fig. 3** Machine leaning techniques

As mentioned in the above Table 1 and Fig. 3 the machine learning models investigated for deepfake detection show varied performance depending on feature extraction and classifier combination. Simple models employing GLCM features with Random Forest or SVM attained relatively low accuracies of 55.63% and 56.8% respectively, showing that low-level texture features are not adequate for consistent classification.

An independent SVM classifier worked slightly better (59.53%), but the actual improvement was obtained using HOG features, with Random Forest + HOG working at 75.58%. This indicates that gradient direction information greatly aids the model in recognizing manipulations.

Additional performance improvement was realized using ensemble models. Stacked ensemble of Random Forest, XGBoost, and SVM gave 76%, while voting ensemble of HOG + XGBoost + SVM obtained 84.62%, the best across all ML models tried. The most streamlined pipeline—Forest + XGBoost with HOG, PCA, and hyperparameter tuning—realized 79.87%, validating preprocessing and feature optimization.

Table 2 Deep leaning techniques

Deep learning techniques	
EfficientNet-B3	97.95
InceptionV3 and DenseNet121	90.5
InceptionV3-based with Transfer Learning and Fine-Tuning	91.47
Simple CNN Model	89.59

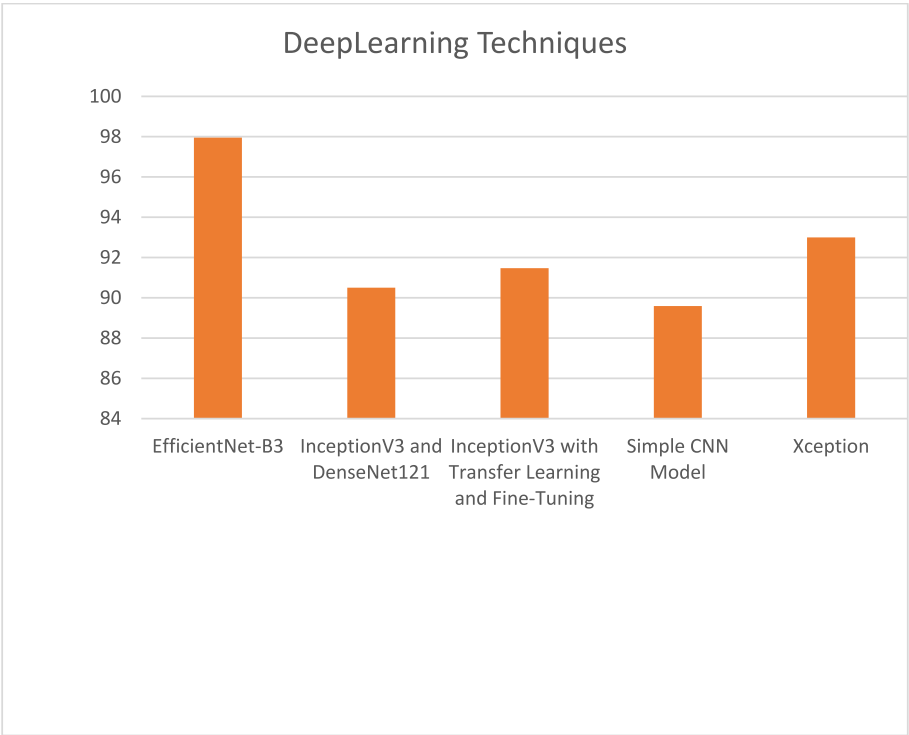


Fig. 4 Deep leaning techniques

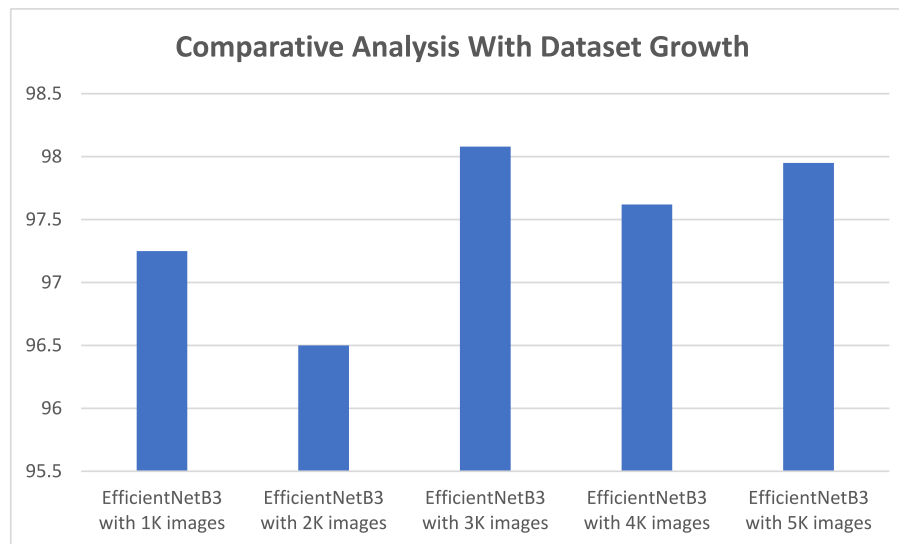
The deep learning models showed much better accuracy than conventional machine learning methods as shown in the above Table 2 and Fig. 4. The EfficientNet-B3 model showed the best performance with 97.95% accuracy, reflecting its strong ability in extracting deep semantic features with computational efficiency. The InceptionV3 and DenseNet121 hybrid model posted 90.5%, reflecting the power of integrating multiple architectures, although still inferior to more sophisticated models.

A dedicated InceptionV3-based deepfake detector, boosted by transfer learning and fine-tuning, marginally increased the performance to 91.47%, indicating that pre-trained weights on big datasets can enhance detection accuracy. The Simple CNN model performed decently at 89.59%, emphasizing the weakness of shallow architectures in modeling intricate forgery patterns.

As mentioned in Table 3 the EfficientNet-B3 model performance was measured with different sizes of datasets varying from 1000 to 5000 images. The model achieved an accuracy rate of 97.25% using 1 K images, which shows optimal performance even in minimal data scenarios. Surprisingly, a decrement to 96.5% was noted while using 2 K images, possibly because increased complexity in the data without effective representation balance causes this (Fig. 5).

Table 3 Comparative analysis with dataset growth

comparative analysis with dataset growth	
EfficientNetB3 with 1 K images	97.25
EfficientNetB3 with 2 K images	96.5
EfficientNetB3 with 3 K images	98.08
EfficientNetB3 with 4 K images	97.62
EfficientNetB3 with 5 K images	97.95

**Fig. 5** Comparative analysis with dataset growth

As the dataset grew to 3 K images, accuracy reached a maximum of 98.08%, indicating the improved learning and generalization ability of the model with more and diverse training samples as shown in the above Fig. 3. Scaling up to 4 K and 5 K images caused a slight decline in performance with accuracies of 97.62% and 97.95% respectively, indicating a plateau in gains in model learning. This means that although data augmentation and volume tend to improve performance, gains become marginal after a certain point, highlighting the importance of quality and diversity over raw quantity.

The experimental results quite evidently indicate that deep learning models by far surpass conventional machine learning methods in detecting DeepFakes. ML models incorporating GLCM features with Random Forest and SVM produced low accuracies of 55.63% and 56.8%, respectively, whereas performance improved moderately with HOG features and better still with ensemble methods—giving a high of 84.62% using a voting ensemble of HOG, SVM, and XGBoost. These findings place emphasis on the weaknesses of primitive texture descriptors and the strength of feature-dense, gradient-descent-based methodologies when paired with an ensemble of classifiers. Alternatively, deep architectures such as EfficientNet-B3, which illustrated greater ability through a 97.95% accuracy rate from learning deep semantic features, exhibited greater capability than fine-tuned InceptionV3 (91.47%), which each also performed impressively, illustrating the benefit in utilizing deeper frameworks and transfer learning. Also, trying EfficientNet-B3 with different dataset sizes showed that even with 1,000 images, the model still had good accuracy (97.25%), reaching a high of 3,000 images (98.08%) before experiencing marginal gains with very large datasets. This demonstrates that the quality and

diversity of data are more important than sheer numbers. Overall, although ML models provide interpretability and reduced complexity, deep learning models offer much higher accuracy and resilience for scalable and reliable DeepFake detection.

On the additional test set comprising AI-generated images from ChatGPT and Gemini, the proposed model successfully classified every image as fake, yielding a detection accuracy of 100%. This outcome highlights the model's strong generalization capability to unseen generative content and reinforces its robustness for practical deployment.

6 Model architecture

The suggested DeepFake detection architecture is based on a deep learning-driven framework that takes advantage of the EfficientNet-B3 model as the backbone as shown in Fig. 6. This framework is crafted to harness the capability of transfer learning, smart data augmentation, and fine-tuning towards high-precision identification of manipulated images. The model pipeline incorporates data preprocessing, semantic feature extraction via deep convolutional layers, and binary classification via a custom classifier head optimized for the DeepFake detection task (Fig. 7).

- Data Preprocessing and Augmentation
 - Input imagery are initially resized to 224×224 pixels to provide uniformity in input dimensions and compatibility with the pretrained EfficientNet-B3 architecture. Random horizontal flip, random rotation, and normalization (with regard to ImageNet mean and standard deviation) are performed during training for enhancing model generalization and robustness. Test and validation set only experience resizing and normalization in order to preserve consistency while evaluation. This two-pronged approach ensures that the model is trained on diverse inputs but evaluated on clean, standardized inputs.

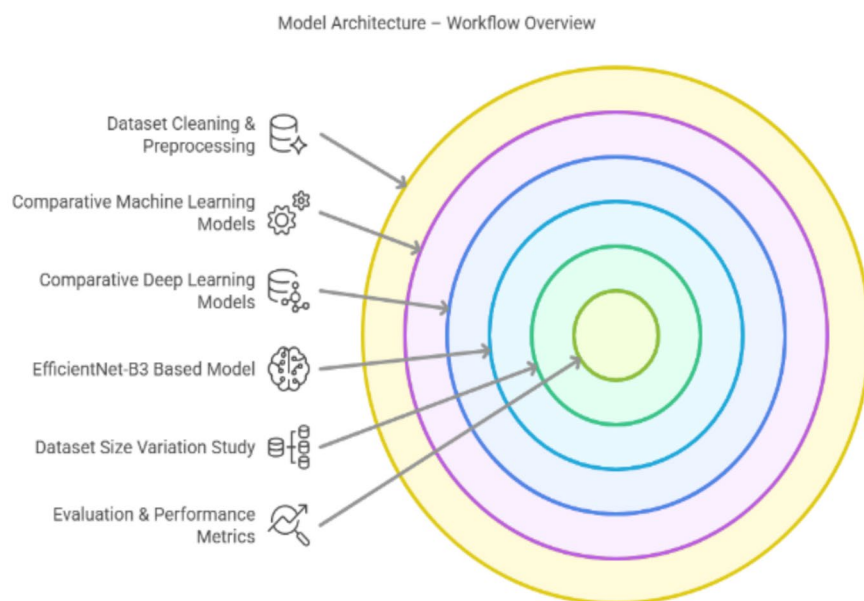


Fig. 6 Model architecture—workflow overview

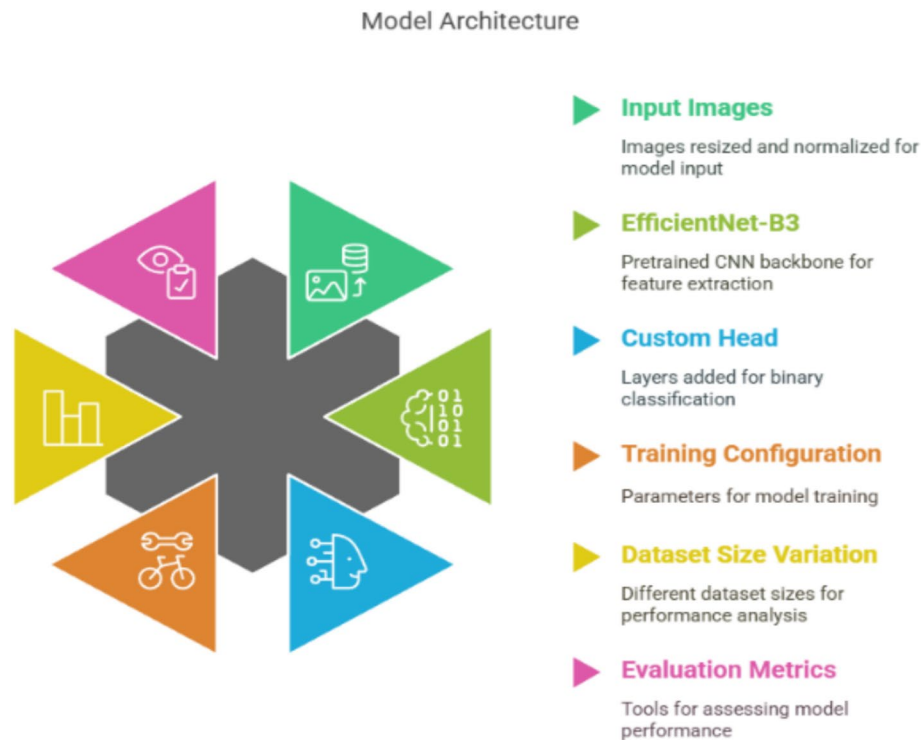


Fig. 7 Model architecture EfficientNet-B3

- Deep Feature Extraction Using EfficientNet-B3
 - o At the center of the architecture is the EfficientNet-B3 model, chosen due to its compound scaling efficiency and top performance on image classification benchmarks. It employs a sequence in terms of depthwise independent convolutions along with the squeeze-and-excitation modules to achieve high-level semantic features while preserving computational efficiency. The model is initialized with ImageNet pretrained weights, enabling it to transfer general visual representations into the DeepFake detection application.
 - o At initial training stages, lower layers are frozen to keep pretrained low-level filters. Deep layers are only unfrozen for fine-tuning in order to enable adaptation over task-specific features—like facial texture irregularities and pixel-based inconsistencies caused by DeepFake generating processes.
- Custom Classification Head
 - o To fine-tune the model for binary classification (real/fake), a default classification layer is substituted by a custom head that includes:
 - o A fully linked layer downscaling features to 128 neurons, with ReLU activation,
 - o A Dropout layer (rate=0.3) to counteract overfitting,
 - o A second fully linked layer with 64 neurons and the RELU activation function.
 - o Another Dropout layer (rate = 0.2),
 - o An output layer with an single neuron with a Sigmoid activation function, outputting probabilities for binary decision-making. This light and compact classifier head is

expressive and lightweight, capable of accurate binary classification without adding too many parameters.

- Training Strategy and Optimization
 - o The model is trained using the Binary Cross-Entropy Loss function, which is suitable for two-class problems. The Adam optimiser is selected, which adapts the learning rates for each weight, allowing for quicker and more stable convergence. The learning rate is set to 0.0001, and the model is taught for 10 epochs plus a batch size of 8. During training, model monitors its performance in terms of accuracy and loss, and the best weights are saved based on validation results.
- Scalability and Dataset Size Analysis
 - o To evaluate scalability, the model is trained and tested on varying dataset sizes—1 K, 2 K, 3 K, 4 K, and 5 K images per class. Despite smaller datasets (1 K) showing impressive results (97.25%), peak performance is reached with 3 K images (98.08%), and larger datasets provide marginal gains. This analysis reveals the model's strong capacity to generalize and emphasizes the importance of data quality and diversity over sheer volume.
- Model Evaluation
 - o Post-training, the model is tested through accuracy, precision, recall, F1-score, and the confusion matrix to check categorization performance exhaustively. Test is done on an independent internal test split (20%) to provide an unbiased validation. The model has consistent high performance with a highest accuracy of 98.08%, validating the architecture's performance in identifying DeepFake images.
 - o This architecture exhibits a strong combination of transfer learning, deep convolutional feature extraction, and task-specific fine-tuning. With excellent generalization, efficiency, and insensitivity to visual forgery artifacts, EfficientNet-B3 is an extremely effective model for real-world DeepFake detection applications.
 - o Apart from typical benchmark datasets, the work also included the process of creating additional test set of images generated by the current AI image generators (ChatGPT/DALL-E and Google Gemini). The images were not seen during the training process and were employed only for testing purposes of the proposed framework for generalization of the model's ability.

7 Algorithm explanation

The developed DeepFake detection system is created based on a fine-tuned EfficientNet-B3 architecture that utilizes transfer learning to determine images as real or fake. The algorithm uses a systematic sequence of operations aimed at maximizing training effectiveness and detection quality as represented in Fig. 8.

Step 1: Loading and preprocessing of data.

- Real and fake image datasets are loaded from distinct directories.
- The images are resized to 224×224 pixels to align with the requirement of EfficientNet-B3.

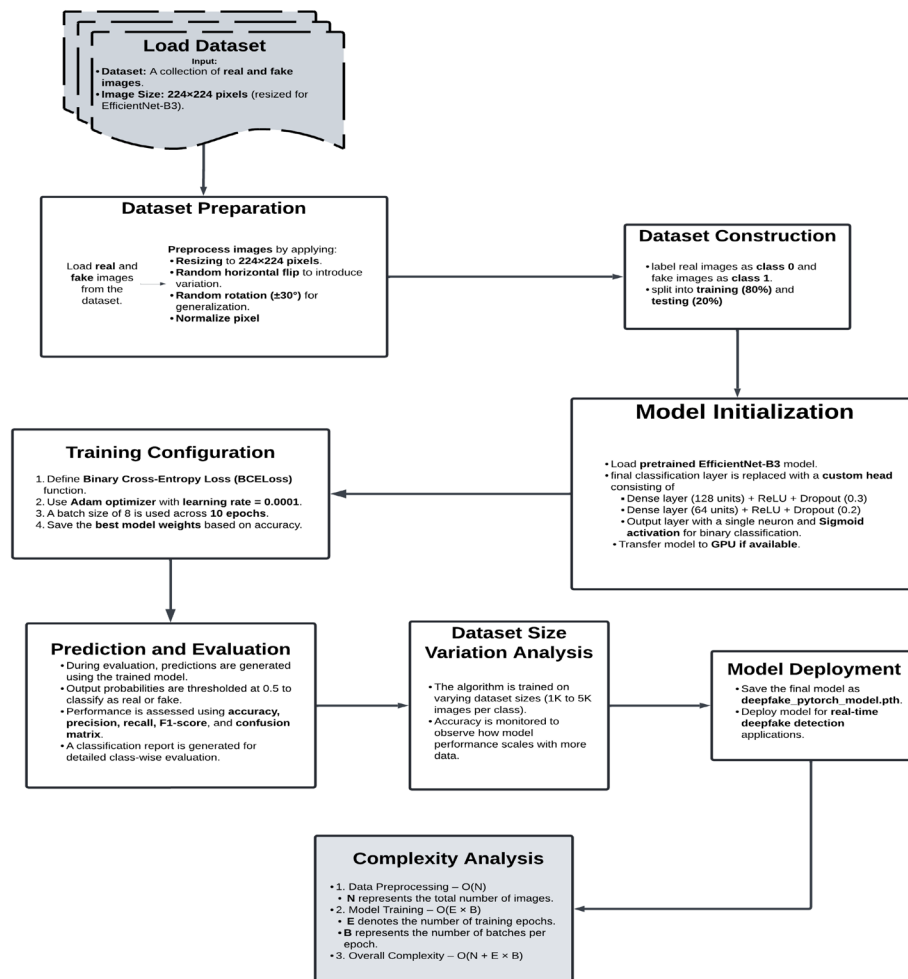


Fig. 8 Workflow of the model

- Data augmentation techniques like random horizontal flip and the rotation are used to add diversity to training data.
- The images are normalized using the mean and standard deviation of ImageNet.

Step 2: Building the dataset.

- A dataset class is built specially to mark real images as class 0 and fake images as class 1.
- The dataset is split into 80% for guidance and 20% for examining using stratified sampling to preserve class balance.

Step 3: Model architecture.

- The base model is EfficientNet-B3 pretrained on ImageNet.
- The classification layer is overridden with a custom head of:
 - Fully connected layer with 128 units, followed by ReLU activation and a dropout rate of 0.3
 - Fully connected layer with 64 units, followed by ReLU activation and a dropout rate of 0.2

- o Output unit with a one neuron and the Sigmoid activation for the binary classification.

Step 4: Training configuration.

- Training is carried out using the Binary Cross-Entropy loss function alongside the Adam optimizer, configured with a learning rate of 0.0001
- Batch size of 8 is utilized over 10 epochs.
- The model is executed on GPU (if present) to reduce computation time.
- Best model weights are stored with respect to validation accuracy while training.

Step 5: Prediction and evaluation.

- At the time of evaluation, predictions are made based on the trained model.
- Output probabilities are thresholded at 0.5 to distinguish between authentic and fake instances.
- The model's performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and the confusion matrix
- A classification report is produced for in-depth class-wise analysis.

Step 6: Dataset size variation analysis.

- The algorithm is trained on datasets of different sizes (1 K to 5 K images per class).
- Accuracy is tracked to see how well model performance increases with increased data.
- The model performed best at 3 K images (98.08%), showing the best training size.

This algorithm balances efficient feature extraction, well-balanced training, and efficient evaluation, and therefore works best for real-world DeepFake detection scenarios.

8 Mathematical representation of the model

The proposed DeepFake detection system is a binary classification model based on EfficientNet-B3 architecture. This section presents the mathematical representation of its core components, from input transformation to prediction and optimization.

8.1 Data transformation

- **Resize:** The image is resized to the fixed size (224, 224) to standardize the input.
- $\text{Resize}(I) \rightarrow I_{\text{resized}}$, where $I \in \mathbb{R}^{H \times W \times C}$

Here, H and W represent the image's height and width respectively, while I denotes the number of color channels. (3 for RGB).

- **Random Horizontal Flip:** This is a data augmentation technique where the image has a 50% chance of being flipped horizontally.

$$\text{Flip}(I) \rightarrow I_{\text{flipped}}, \text{ with probability } p = 0.5$$

- **Random Rotation:** Random rotation by an angle up to 30 degrees.

$$\text{Rotate}(I, \theta) \rightarrow I_{\text{rotated}}, \theta \in \mathbb{R}, |\theta| \leq 30^\circ$$

- Normalization: Pixel values are scaled based on the mean and standard deviation derived from the ImageNet dataset.

$$\text{Normalize}(I) = \frac{I - \mu}{\sigma}$$

where $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.299, 0.244, 0.225]$ represent the mean and standard deviation for each individual image channel.

8.2 Model architecture: EfficientNet-B3

The model used is EfficientNet-B3, a neural network architecture designed for efficiency with fewer parameters while achieving high performance. It utilizes weights learned from training on ImageNet and then modified for binary classification (real vs fake). The classifier portion of the model is as follows:

$$\text{Output} = \text{Sigmoid}(\text{Linear}(\text{ReLU}(\text{Linear}(x))))$$

where:

Linear Layer 1: $\text{Linear}(x) \rightarrow x'$ with output size 128.

ReLU: Applies activation function, $\text{ReLU}(x') = \max(0, x')$

Dropout: A regularization technique to prevent the overfitting cases by randomly deactivating certain neurons during training.

Linear Layer 2: $\text{Linear}(x') \rightarrow x''$ with output size 64.

ReLU: Again applies ReLU activation.

Dropout: Another regularization layer.

Linear Layer 3: Outputs the final predictions for binary classification with output size 1.

Final Output = $\text{Sigmoid}(x'')$ where $\text{Sigmoid}(z) = \frac{1}{1+e^{-z}}$

This produces a probability $p \in [0,1]$ for each input image, where the output is interpreted as the probability of image being fake.

8.3 Loss function: binary cross-entropy loss

The model is trained using the binary cross-entropy loss function. The loss is calculated as follows:

$$L = -\frac{1}{N} \sum_i \log(p_i^{y_i}) = -\frac{1}{N} \sum_i [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where:

N , number of the samples in a batch.

y_i true label (0 or 1) for the sample i .

p_i is predicted as the probability that the sample is fake.

8.4 Optimizer: Adam

Adam optimizer is employed to minimize the model's loss function. It is an adaptive optimization algorithm that adjusts the learning rate. Update rule for Adam is as follows:

$$\theta_t = \theta_{t-1} - \frac{\alpha}{\sqrt{v_t} + \epsilon} m_t$$

where:

α is the learning rate (0.0001).

m_t and v_t are the first and the second moment estimates (mean and variance of the gradients).

ε is a small constant to prevent division by zero.

8.5 Training loop

- For each epoch, the model undergoes the following steps:
- Forward Pass: Compute the model's output for each batch.

$$\text{output} = \text{model}(x)$$

Loss Calculation: Compute the loss for the batch.

$$\text{loss} = L(\text{output}, y)$$

Backward Pass: Compute gradients for all parameters.

$$\nabla L(\text{output}, y)$$

Optimizer Step: Update model parameters to minimize the loss.

8.6 Evaluation metrics

This model's performance is evaluated using:

$$\text{Accuracy} = \frac{\text{Number of the total Correct Predictions}}{\text{Total Number of the Predictions}}$$

Classification Report: This includes the precision, recall, and F1-score for each class.

Confusion Matrix: to visualize the performance of the model.

$$\text{Confusion Matrix} = \begin{pmatrix} \text{TP} & \text{FP} \\ \text{FN} & \text{TN} \end{pmatrix}$$

This confusion matrix is employed to calculate:

- **Precision:** $\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$
- **Recall:** $\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$
- **F1-Score:** $\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

8.7 Summary of the mathematical structure:

- Data Preprocessing: Image resizing, augmentation, and normalization.
- Model Architecture: EfficientNet-B3 with additional fully connected layers for binary classification.
- Loss Function: Binary cross-entropy.
- Optimization: Adam optimizer.
- Metrics: Accuracy, precision, recall, F1-score, and confusion matrix for evaluation.

9 Results and discussion

The proposed DeepFake detection framework is built upon the EfficientNet-B3 backbone, leveraging its ability to capture complex visual patterns with fewer parameters and computational cost. During training, the model was fine-tuned using pre-trained ImageNet weights, followed by a staged unfreezing and optimization of deeper layers.

Training was conducted for 10 epochs, during which the model exhibited consistent improvements in both accuracy and loss, highlighting effective feature learning and model convergence. The learning trajectory can be summarized as:

- Epoch 1: Accuracy: 87.72%, Loss: 0.2827
- Epoch 5: Accuracy: 98.38%, Loss: 0.0431
- Epoch 10: Accuracy: 99.39%, Loss: 0.0166

This steady drop in loss and rise in accuracy demonstrates the model's strong learning capability, culminating in high confidence predictions by the end of training. The integration of data augmentation, dropout layers, and fine-tuning strategies helped in achieving this high performance while minimizing overfitting.

On evaluation using the test dataset, the EfficientNet-B3 model achieved an excellent accuracy of 97.95%, establishing its effectiveness for DeepFake detection tasks.

When evaluated on this supplementary set, the model correctly identified all AI-generated images as fake, achieving 100% detection accuracy on ChatGPT and Gemini outputs. This additional evaluation demonstrates that the proposed model generalizes effectively to unseen AI-generated content, further validating its robustness and practical applicability.

9.1 Classification report

The model was tested on a total of 2000 images, equally distributed across real and fake classes (991 real, 1009 fake). The evaluation metrics obtained are as follows:

- Precision: Indicates how many of the predicted real/fake labels were correct.
- Recall: Processes the capability of this model to accurately recognize actual real/fake images.
- F1-score: Represents the harmonic average of precision and recall

All metrics hover around 0.98 as shown in the above Fig. 9 and Table 4, showcasing a balanced and accurate performance across both classes. The high F1-score confirms the model's strong predictive capability in both detecting fake faces and verifying real ones.

9.2 Confusion matrix analysis

The confusion matrix serves as the powerful tool which is used for assessing the classification performance of these models by demonstrating the relationship between true and predicted classes. In the context of this research, it provides valuable insights into how effectively the EfficientNet-B3 model differentiates between real and DeepFake facial images as shown in Fig. 10.

9.2.1 Matrix overview

- True Positives (TP)=971: These are the real images that the model accurately categorized as real.

Classification Report:				
	precision	recall	f1-score	support
Real	0.98	0.98	0.98	991
Fake	0.98	0.98	0.98	1009
accuracy			0.98	2000
macro avg	0.98	0.98	0.98	2000
weighted avg	0.98	0.98	0.98	2000

Fig. 9 Classification report

Table 4 Classification reports

Class	Precision	recall	F1-score	Support
Real	0.98	0.98	0.98	991
Fake	0.98	0.98	0.98	1009
Overall accuracy			0.9795	2000

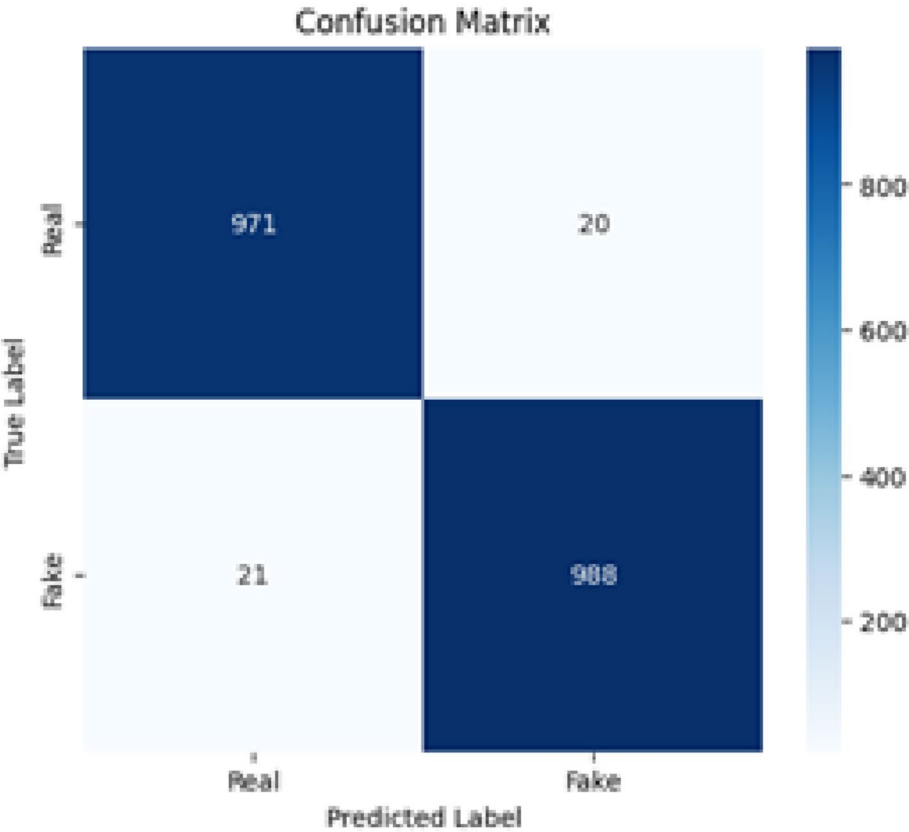


Fig. 10 Confusion matrix

- True Negatives (TN) = 988: These are the fake images that the model accurately identified as fake.
- False Positives (FP) = 21: These fake images were incorrectly classified as real, posing a potential risk in security-sensitive environments.
- False Negatives (FN) = 20: These real images were misclassified as fake, which could be problematic in applications where legitimate content must not be flagged.

9.3 Interpretation and impact

9.3.1 High true positive and true negative rates

- o The large values of TP and TN indicate that the model has learned strong decision boundaries between the two classes.
- o With 971 out of 991 real images and 988 out of 1009 fake images correctly identified.

9.3.2 Low false positives

- o The 21 false positives represent fake images that were mistakenly classified as real. This type of error can have significant implications in real-world applications, such as misinformation detection, where the primary goal is to prevent deceptive content from being accepted as authentic.
- o However, the relatively low count reflects the model's cautious decision-making in not over-trusting visual cues that may appear genuine.

9.3.3 Low false negatives

- o The 20 false negatives refer to real images flagged incorrectly as fake. While less dangerous from a security perspective, such errors can reduce trust in the system, especially if used in user-facing verification platforms.
- o The near symmetry in FP and FN (21 vs. 20) also demonstrates that the model maintains a balanced bias between the two classes, avoiding favouring one category over the other.

9.4 Overall accuracy and error rate

- Total Correct Predictions = 971 (TP) + 988 (TN) = 1959
- Total Incorrect Predictions = 20 (FN) + 21 (FP) = 41
- Overall Accuracy = $1959 / 2000 = 97.95\%$
- Error Rate = $41 / 2000 = 2.05\%$

This confirms that only a small fraction of samples was misclassified, reinforcing the model's robustness and suitability for DeepFake detection in practical environments.

9.5 Insights and conclusions

- The near-perfect diagonal in the confusion matrix reflects the strong generalization ability of the EfficientNet-B3 architecture when fine-tuned for binary image classification.
- The model handles both real and fake instances with almost equal proficiency, which is critical for avoiding biased decisions.

- Given the sensitive nature of DeepFake detection, especially in fields like digital forensics, news verification, and biometric authentication, maintaining this balance is vital.
- Further reductions in false positives and false negatives may be achieved by incorporating attention mechanisms, multi-scale feature fusion, or adversarial training techniques.

In conclusion, the proposed method not only outperforms many conventional approaches but also provides scalable and a efficient solution for the practical applications in digital media verification and forensic analysis.

9.6 Comparative performance with other machine learning and deep learning techniques

To validate superiority of proposed EfficientNet-B3 model for DeepFake detection, an extensive comparative analysis was conducted against variety of machine learning (ML) and deep learning (DL) techniques. The objective of this evaluation is to demonstrate how well the proposed framework performs in contrast to both traditional handcrafted approaches and other deep learning-based solutions.

9.7 Deep learning-based models

Deep learning models have achieved significant success in image classification tasks, owing to their capability to learn hierarchical feature representations directly from raw pixel values. The following models were evaluated under similar conditions using pre-processed DeepFake datasets (Table 5).

The EfficientNet-B3 model outperforms all other DL models, delivering a remarkable accuracy of 97.95%. Its architecture benefits from compound scaling and efficient parameter usage, enabling deep and wide networks with fewer resources. Even high-performing architectures such as InceptionV3 and DenseNet121 fall short by over 7% in accuracy, highlighting the strength of EfficientNet-B3 in generalizing complex patterns found in manipulated facial images.

9.8 Machine learning-based models

Several traditional machine learning techniques were also compared using handcrafted Features such as the Gray Level Co-occurrence Matrix (GLCM) and Histogram of Oriented Gradients (HOG). These methods generally require careful feature engineering and data preprocessing to achieve optimal results (Table 6).

Among ML models, the HOG + XGBoost + SVM ensemble yields the best performance at 84.62%, showcasing the benefit of combining classifiers. However, even this carefully tuned ensemble lags behind deep learning models by a considerable margin. Traditional models also struggle to generalize well to complex facial forgery patterns, especially in the presence of subtle manipulations.

Table 5 Deep learning based model's

Deep learning technique	Accuracy (%)
Proposed EfficientNet-B3 model	97.95
Inceptionv3 and densenet121 combination	90.50
Inceptionv3-based detector with transfer learning and fine-tuning	91.47
Simple convolutional neural network (CNN)	89.59

Table 6 Machine learning-based models

Machine learning technique	Accuracy (%)
GLCM + random forest	55.63
SVM + GLCM	56.80
SVM classifier (raw features)	59.53
Random forest with hog features	75.58
Stacked ensemble (random forest + XGBOOST + SVM, META: logistic regression)	76.00
HOG + XGBOOST + SVM (soft voting ensemble)	84.62
Random forest + XGBOOST (with preprocessing, HOG, PCA, and hyperparameter tuning)	79.87

Table 7 Comparative analysis of existing deepfake identification models

Author(s)	Model used	Accuracy (%)
Sohail Ahmed Khan et al. (2021)	Fusion of VGG16, InceptionV3, and XceptionNet	96.5
Luca Guarnera et al. (2022)	Encoder trained on latent space of multiple GAN models	96.0
Intel (2022)	FakeCatcher (Real-time Detection via Blood Flow Analysis)	96.0
Dereesa Wodajo et al. (2023)	Generative Convolutional Vision Transformer (GenConViT)	95.8
Tianchen Zhao et al. (2020)	Pair-wise Self-Consistency Learning (PCL)	98.05
Varun et al. (2022)	GAN with CNN-based DeepFake Detection	95.86
Hao et al. (2022)	CNN-based Approach on DeepFake Detection Dataset (DFDC)	91.88
Rushit Dave & Mounika Vanamala (2022)	CNN and SVM Hybrid Model	88.33

10 Comparative analysis of existing deepfake identification models

DeepFake detection has become a pivotal research field as a result of its impact on media authenticity, disinformation, and online security. Multiple models have been proposed to identify authentic and artificially produced facial images using diverse architectural designs and learning patterns. This section offers a comparative summary of the current DeepFake discovery techniques based on their underlying methodologies, model architectures, and classification accuracy.

As mentioned in the above Table 7 some of the Deepfake detection models have shown excellent performance based on various methodologies. Models based on fusion, such as that of Sohail Ahmed Khan et al. (2021), were able to detect with 96.5% accuracy by fusing VGG16, InceptionV3, and XceptionNet. Guarnera et al. (2022) and Intel (2022) also achieved 96.0%, using GAN-based encoders and physiological signals, respectively. Transformer-based models such as GenConViT by Wodajo et al. (2023) also performed well (95.8%), while Zhao et al. (2020) achieved the highest reported accuracy (98.05%) with pair-wise consistency learning.

Other models such as GAN-enhanced CNNs (Varun et al., 2022) and hybrid CNN-SVM methods (Dave & Vanamala, 2022) achieved 95.86% and 88.33%, respectively. These findings indicate that ensemble learning, and architectural novelty are the drivers of high accuracy.

Our suggested EfficientNet-B3 based model continues this trend by employing transfer learning and fine-tuning to achieve an accuracy of 97.95%. It strikes a balance between depth, efficiency, and generalization, making it a very competitive solution in the Deep-Fake detection space.

11 Performance evaluation of the model

Performance of proposed DeepFake discovery model, built upon the EfficientNet-B3 architecture, was estimated employing standard classification metrics as shown in Figs. 9 and 10. These metrics grant a complete through assessment of the model's ability to recognize between real and synthetically manipulated facial images.

11.1 Accuracy and generalization

It achieved an outstanding test accuracy of 97.95% on balanced test set of 2000 images (991 real and 1009 fake). This high accuracy reflects this model's robust feature extraction and classification capabilities, bolstered by transfer learning and effective data augmentation strategies. The consistent rise in accuracy across training epochs, culminating in 99.39% training accuracy, indicates stable learning without signs of overfitting.

11.2 Precision, recall, and F1-score

The report demonstrated near-identical scores across both classes:

- Precision: 0.98 for both real and fake classes
- Recall: 0.98 for both classes
- F1-Score: 0.98 for both classes

These results highlight the model's balanced performance, where both false positives and false negatives are minimized, ensuring reliable classification across different types of image manipulations.

11.3 Confusion matrix analysis

The confusion matrix showed:

- 971 True Positives (real images correctly classified),
- 988 True Negatives (fake images correctly classified),
- Only 20 False Negatives and 21 False Positives.

This low misclassification rate further validates the model's discriminative strength and consistency.

11.4 Training performance over epochs

The training process over 10 epochs demonstrated a steady reduction in loss from 0.2827 in epoch 1 to 0.0166 in epoch 10, while accuracy improved from 87.72% to 99.39%. This trend reflects efficient convergence and the effectiveness of the optimizer, loss function, and learning rate settings.

11.5 Scalability and data size sensitivity

Experiments with varying dataset sizes (1 K to 5 K per class) showed that EfficientNet-B3 maintained high accuracy even with limited data (97.25% at 1 K), peaking at 98.08% with 3 K images. This demonstrates strong generalization and adaptability to both small- and large-scale training scenarios.

12 Conclusion

In the proposed work, a DeepFake detection model was developed using the EfficientNet-B3 architecture, supported by transfer learning, data augmentation, and fine-tuning techniques. The model demonstrated strong performance, achieving a test accuracy of 97.95%, with balanced precision and recall across real and fake image classes. The training behavior reflected stable convergence, and the results indicate that the model can handle variations in dataset size with reasonable consistency. The model also correctly classified a small supplementary set of AI-generated images from ChatGPT and Gemini, suggesting its potential to recognize patterns from emerging generative technologies, though broader testing is required to confirm this.

A comparative analysis with deep and machine-learning baselines showed that the proposed method performed better than traditional handcrafted-feature approaches and remained competitive with several deep learning-based detectors evaluated in this study. Its architectural efficiency also makes it a practical candidate for applications where computational resources are limited.

Overall, the findings indicate that EfficientNet-B3, when appropriately optimized, can serve as a capable solution for DeepFake image classification. However, additional evaluation across diverse datasets, newer generative models, and video-based DeepFakes is needed to fully establish its generalizability for real-world deployment.

13 Future improvements

While the suggested EfficientNet-B3 model performed well in DeepFake detection, there are avenues for improvement. Future research can investigate expanding the model to video-based DeepFakes by adding temporal features. Enhancing robustness against adversarial attacks and compressing the model for real-time execution on edge devices are also potential avenues. Testing the model on various datasets can also enhance its generalization in real-world applications.

14 Limitations of the study

Although the proposed EfficientNet-B3 model performed well in image-level DeepFake detection, several limitations should be acknowledged. First, the experiments were conducted on a single publicly available dataset, and the model's performance may vary across other DeepFake datasets with different characteristics. Second, the evaluation focused only on still images; video-based DeepFakes, which contain temporal artifacts, were not examined in this study. Third, the supplementary AI-generated images used to test generalization were limited in number, and broader validation across more generative models is required to draw stronger conclusions. In addition, adversarial robustness, compression effects, and other real-world distortions were not explicitly evaluated. Finally, model efficiency on edge devices and real-time environments was not assessed, which is an important consideration for practical deployment.

Authors contribution

BGD contributed to the conceptualization of the research, and wrote the Abstract, Introduction, and Significance and Contribution sections. CKL conducted the Review of Literature and supported the Comparative Analysis of Existing DeepFake Identification Models. DU was responsible for Dataset collection, Preprocessing, and the implementation of the Model Architecture. BA developed the Algorithm Explanation and formulated the Mathematical Representation of the Model. PVY performed the Results and Discussion, Performance Evaluation, and contributed significantly to the Result Analysis. KPS contributed to the Conclusion, final manuscript proofreading, and coordinated the manuscript formatting and submission process. All authors participated in revising the manuscript critically, approved the final version, and agreed to be accountable for all aspects of the work.

Funding

No funding received by any government or private concern.

Data availability

The datasets analyzed during the current study are available in the DeepFake and Real Images repository on Kaggle (<https://www.kaggle.com/datasets/manjilkarki/deepfake-and-real-images>). Additional AI-generated image samples used for validation are available from the corresponding author on reasonable request.

Declarations**Ethics approval and consent to participate**

This study uses publicly available datasets and does not involve human participants, personal data, or interventions. Therefore, ethical approval is not required. No human participants were involved in this research.

Consent for publication

Not applicable. No identifiable human data is included in this study.

Competing interests

The authors declare no competing interests.

Received: 15 May 2025 / Accepted: 23 December 2025

Published online: 10 January 2026

References

1. Khan SA, Artusi A, Dai H. Adversarially robust deepfake media detection using fused convolutional neural network predictions. 2021. arXiv preprint [arXiv:2102.05950](https://arxiv.org/abs/2102.05950).
2. Wodajo D, Atnafu S, Akhtar Z. Deepfake video detection using generative convolutional vision transformer. 2023. arXiv preprint [arXiv:2307.07036](https://arxiv.org/abs/2307.07036).
3. Coccomini DA, Messina N, Gennaro C, Falchi F. Combining efficientnet and vision transformers for video deepfake detection. In: International conference on image analysis and processing. Cham: Springer International Publishing; 2022, p. 219–229.
4. Thing VLL. Deepfake detection with deep learning: Convolutional neural networks versus transformers. In: 2023 IEEE international conference on cyber security and resilience (CSR). IEEE; 2023, p. 246–253.
5. Sun F, Zhang N, Xu P, Song Z. Deepfake detection method based on cross-domain fusion. Secur Commun Networks. 2021. <https://doi.org/10.1155/2021/2482942>.
6. Qiu X, Miao X, Wan F, Duan H, Shah T, Ojha V, et al. D2fusion: dual-domain fusion with feature superposition for Deepfake detection. Inf Fusion. 2025;120:103087.
7. Chen H-S, Rouhsedaghat M, Ghani H, Hu S, You S, Jay Kuo C-C. Defakehop: A light-weight high-performance deepfake detector. In: 2021 IEEE international conference on multimedia and expo (ICME). IEEE; 2021, p. 1–6.
8. Wodajo D, Atnafu S. Deepfake video detection using convolutional vision transformer. 2021. arXiv preprint [arXiv:2102.11126](https://arxiv.org/abs/2102.11126).
9. Zhao T, Xu X, Xu M, Ding H, Xiong Y, Xia W. Learning self-consistency for deepfake detection. In: Proceedings of the IEEE/CVF international conference on computer vision. 2021, p. 15023–15033.
10. Passos LA, Jodas D, Costa KAP, Souza Júnior LA, Rodrigues D, Del Ser J, et al. A review of deep learning-based approaches for deepfake content detection. Expert Syst. 2024;41(8):e13570.
11. Tran V-N, Lee S-H, Le H-S, Kwon K-R. High performance deepfake video detection on CNN-based with attention target-specific regions and manual distillation extraction. Appl Sci. 2021;11(16):7678.
12. Rahman A. Deepfake video detection using CNN and RCNN. PhD diss. Bangladesh University of Business and Technology; 2022.
13. Bondi L, Cannas ED, Bestagini P, Tubaro S. Training strategies and data augmentations in cnn-based deepfake video detection. In: 2020 IEEE international workshop on information forensics and security (WIFS). IEEE; 2020, p. 1–6.
14. Heidari A, Jafari Navimipour N, Dag H, Unal M. Deepfake detection using deep learning methods: a systematic and comprehensive review. WIREs Data Min Knowl Discov. 2024;14(2):e1520.
15. Badale A, Castelino L, Darekar C, Gomes J. Deep fake detection using neural networks. In: 15th IEEE international conference on advanced video and signal based surveillance (AVSS). 2018, vol. 2.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com